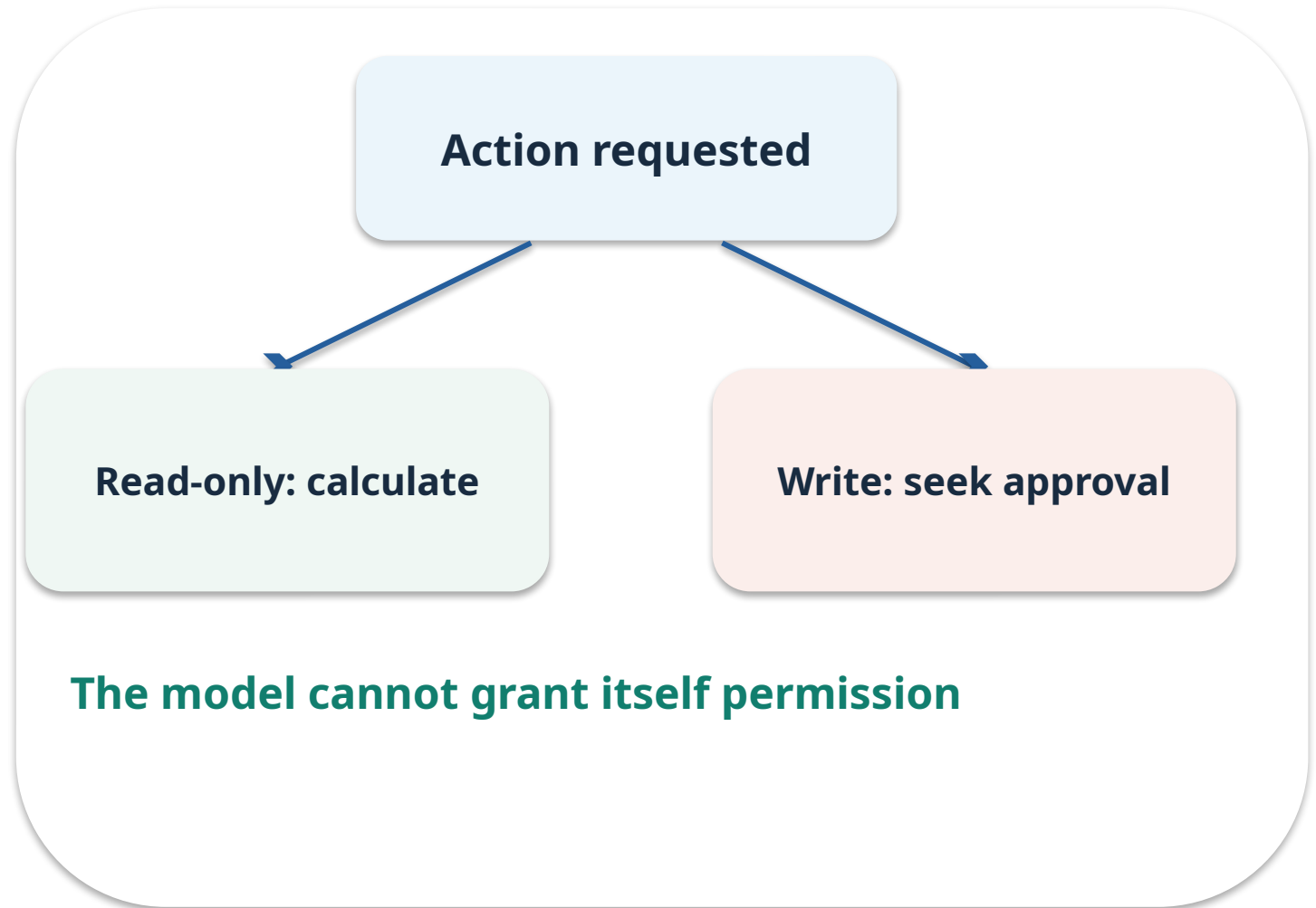


# Week 08 · Tools and controlled actions

## OPEN / WEEK 08

The reimbursement cap tool

A €55 claim has a €40 cap. The assistant may calculate the payable amount. A separate request asks it to change the cap. Which path needs approval?



# First vote: what would you do?

## QUESTION / WEEK 08

A €55 claim has a €40 cap. The assistant may calculate the payable amount. A separate request asks it to change the cap. Which path needs approval?

Choose an initial answer and name one piece of evidence you need.

### VOTE FIRST • GIVE A REASON

**A Accept the first plausible result**

**B Check the source or measurement**

**Which source or test could settle this?**

# The journey through today's case

OPEN / WEEK 08

We will predict, connect an analogy, trace a mechanism, work an example, then explain a decision.



# 1. Predict · Tool

## QUESTION / WEEK 08

What should run before the arithmetic?

Make a choice before we explain the diagram.

### VOTE FIRST · GIVE A REASON

**A Type and range validation.**

**B A tool call is safe because its function is simple.**

**Which source or test could settle this?**

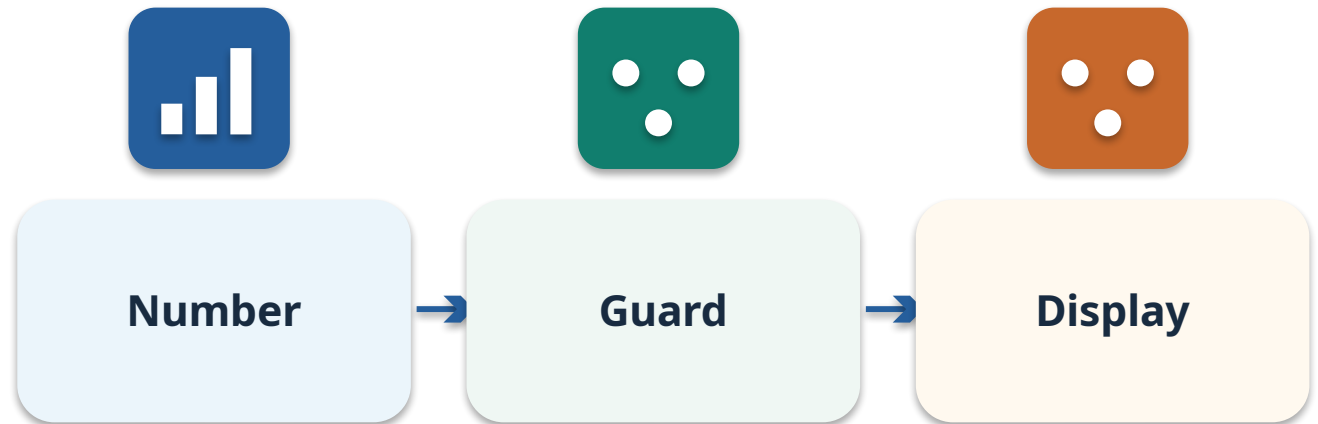
# Tool • a familiar picture

## ANALOGY / WEEK 08

A calculator performs one defined operation when given a valid number.

What matches our case, and where does the analogy stop?

**A calculator performs one defined operation when given a valid number.**



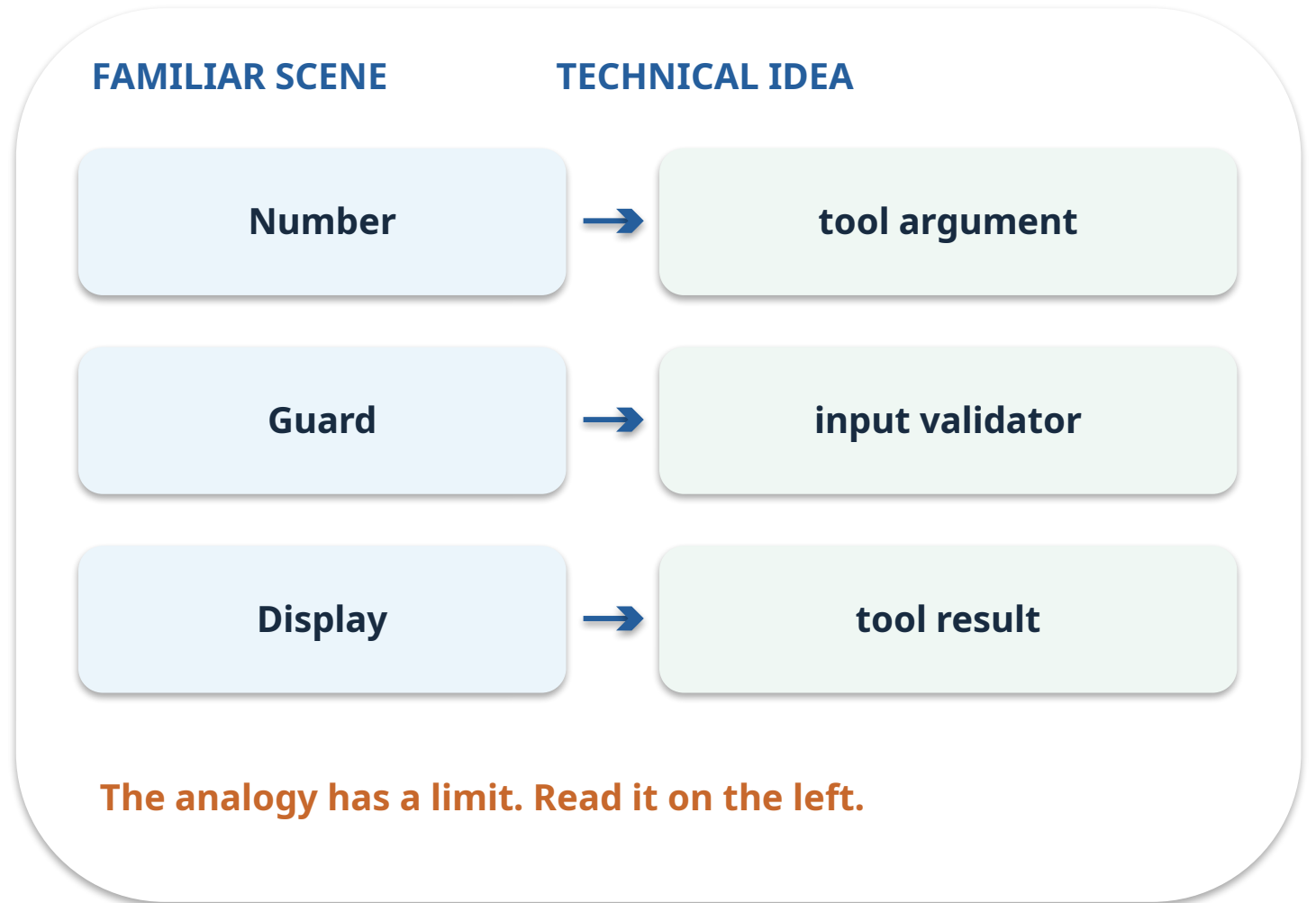
**What matches the technical process?**

# Tool · map the parts

## MAPPING / WEEK 08

Match each everyday part to the technical part.

Analogy limit: The model may choose the wrong tool or arguments; inspect the call.



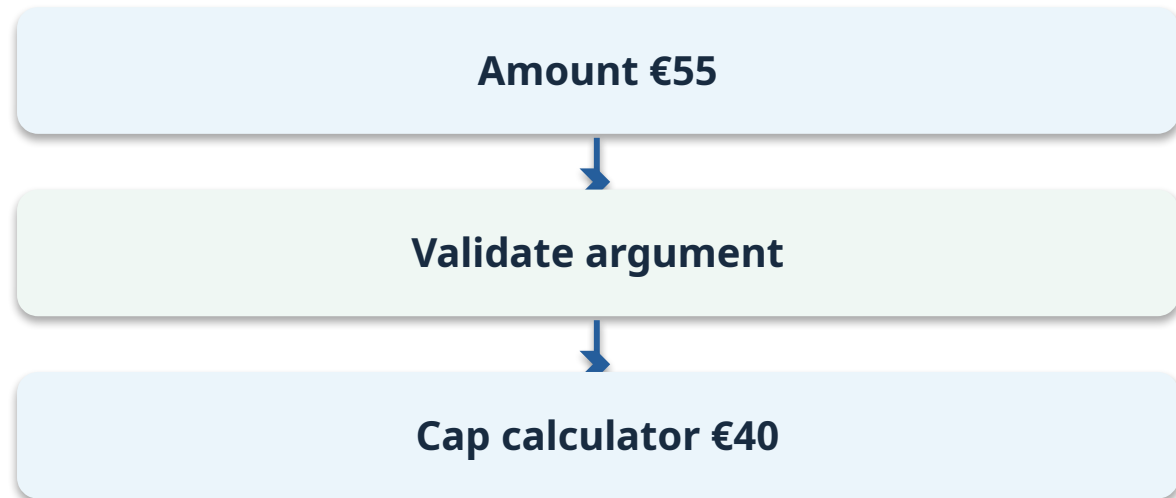
# Tool · how it works

## MECHANISM / WEEK 08

A named operation with validated arguments, permissions and a defined result.

Trace the arrows from left to right.

### FOLLOW THE DATA THROUGH EACH ARROW



# Tool · work one case

## WORKED / WEEK 08

cap(55,40) returns €40;  
cap(-5,40) is rejected as invalid  
input.

Point to the evidence before  
giving your conclusion.

### GIVEN

**cap(55,40) returns €40; cap(-5,40) is rejected as invalid input.**

### QUESTION

**What should run before the arithmetic?**

### SUPPORTED ANSWER

**Type and range validation.**

# Tool • answer and explain

REVEAL / WEEK 08

Type and range validation.

Tempting mistake: A tool call is safe because its function is simple.

## TEST THE CLAIM AGAINST THE EVIDENCE

### TEMPTING CLAIM

**A tool call is safe because its function is simple.**

### SUPPORTED EXPLANATION

**Type and range validation.**

The model may choose the wrong tool or arguments; inspect the call.

## 2. Predict · Workflow

### QUESTION / WEEK 08

Who chooses the next station in this version?

Make a choice before we explain the diagram.

### VOTE FIRST · GIVE A REASON

**A The predefined workflow code.**

**B A workflow must involve a free-form agent.**

**Which source or test could settle this?**

# Workflow · a familiar picture

## ANALOGY / WEEK 08

An assembly line follows the same stations in a fixed order.

What matches our case, and where does the analogy stop?

**An assembly line follows the same stations in a fixed order.**



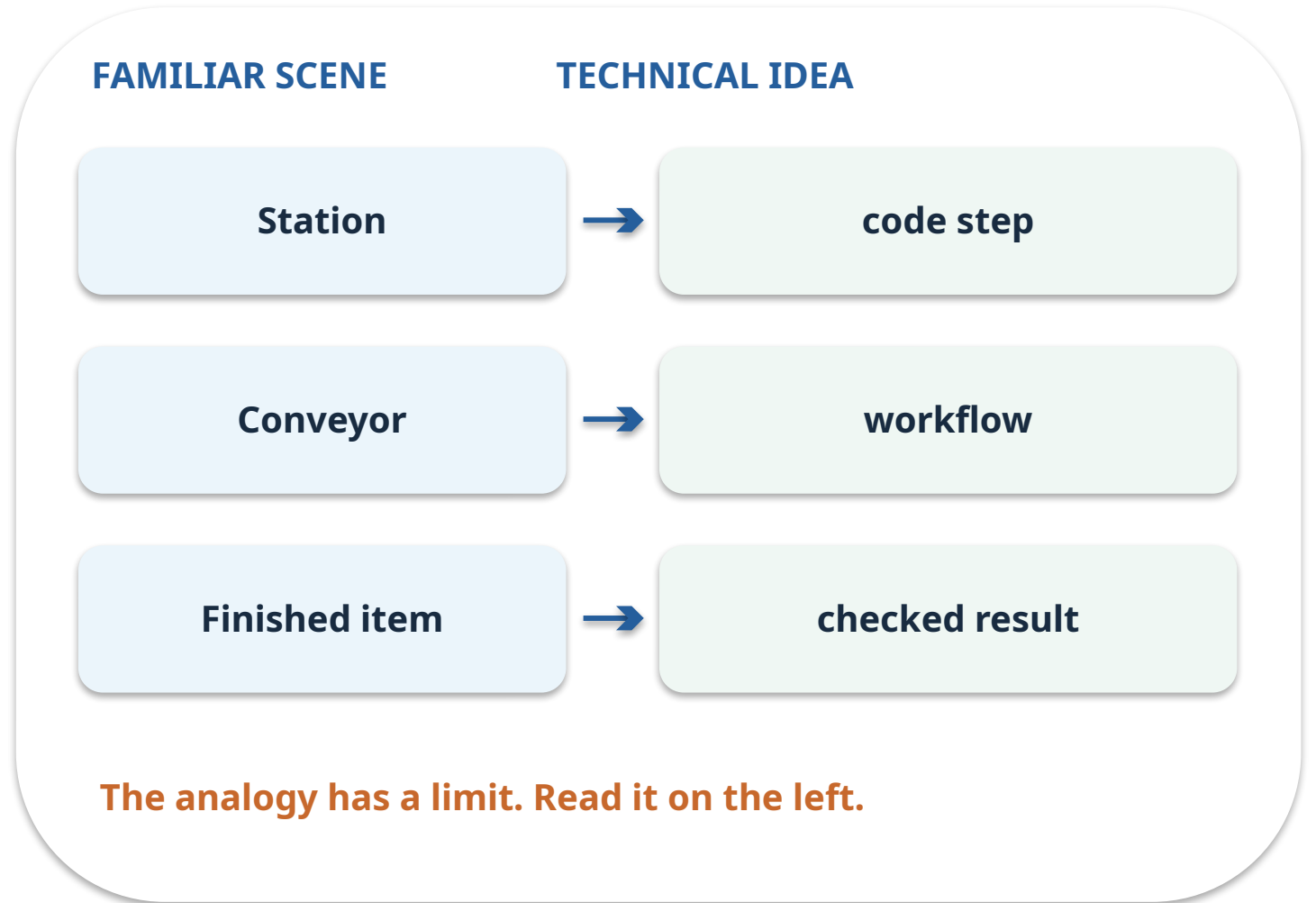
**What matches the technical process?**

# Workflow · map the parts

## MAPPING / WEEK 08

Match each everyday part to the technical part.

Analogy limit: Real workflows can branch on validated conditions, while remaining predefined.



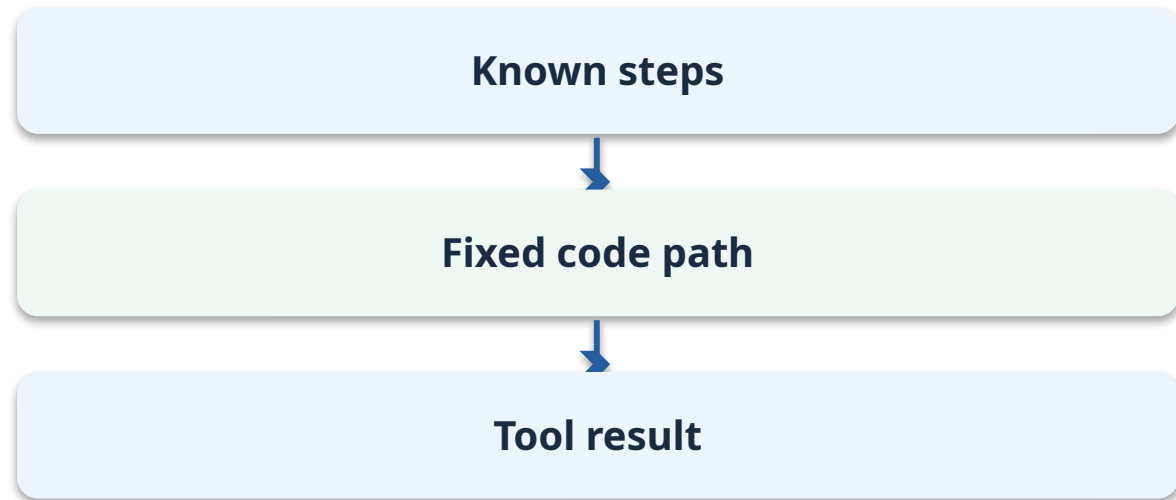
# Workflow · how it works

## MECHANISM / WEEK 08

Code chooses the next step by a fixed rule.

Trace the arrows from left to right.

### FOLLOW THE DATA THROUGH EACH ARROW



# Workflow · work one case

## WORKED / WEEK 08

Validate amount → calculate cap  
→ format reply; a negative  
amount stops at validation.

Point to the evidence before  
giving your conclusion.

### GIVEN

**Validate amount → calculate cap → format reply; a negative amount stops at validation.**

### QUESTION

**Who chooses the next station in this version?**

### SUPPORTED ANSWER

**The predefined workflow code.**

# Workflow · answer and explain

REVEAL / WEEK 08

The predefined workflow code.

Tempting mistake: A workflow must involve a free-form agent.

## TEST THE CLAIM AGAINST THE EVIDENCE

### TEMPTING CLAIM

**A workflow must involve a free-form agent.**

### SUPPORTED EXPLANATION

**The predefined workflow code.**

Real workflows can branch on validated conditions, while remaining predefined.

### 3. Predict · Agent

#### QUESTION / WEEK 08

Can the model authorize a write by choosing a tool?

Make a choice before we explain the diagram.

#### VOTE FIRST · GIVE A REASON

**A No. Authorization is enforced outside the model.**

**B An agent can grant itself a new permission.**

**Which source or test could settle this?**

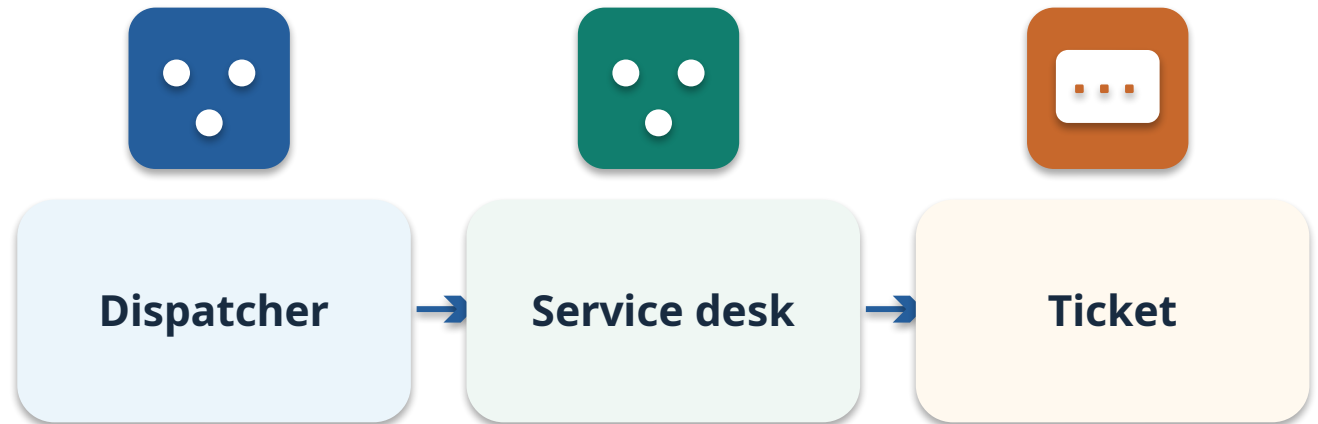
# Agent · a familiar picture

## ANALOGY / WEEK 08

A dispatcher selects among allowed services to handle a request.

What matches our case, and where does the analogy stop?

**A dispatcher selects among allowed services to handle a request.**



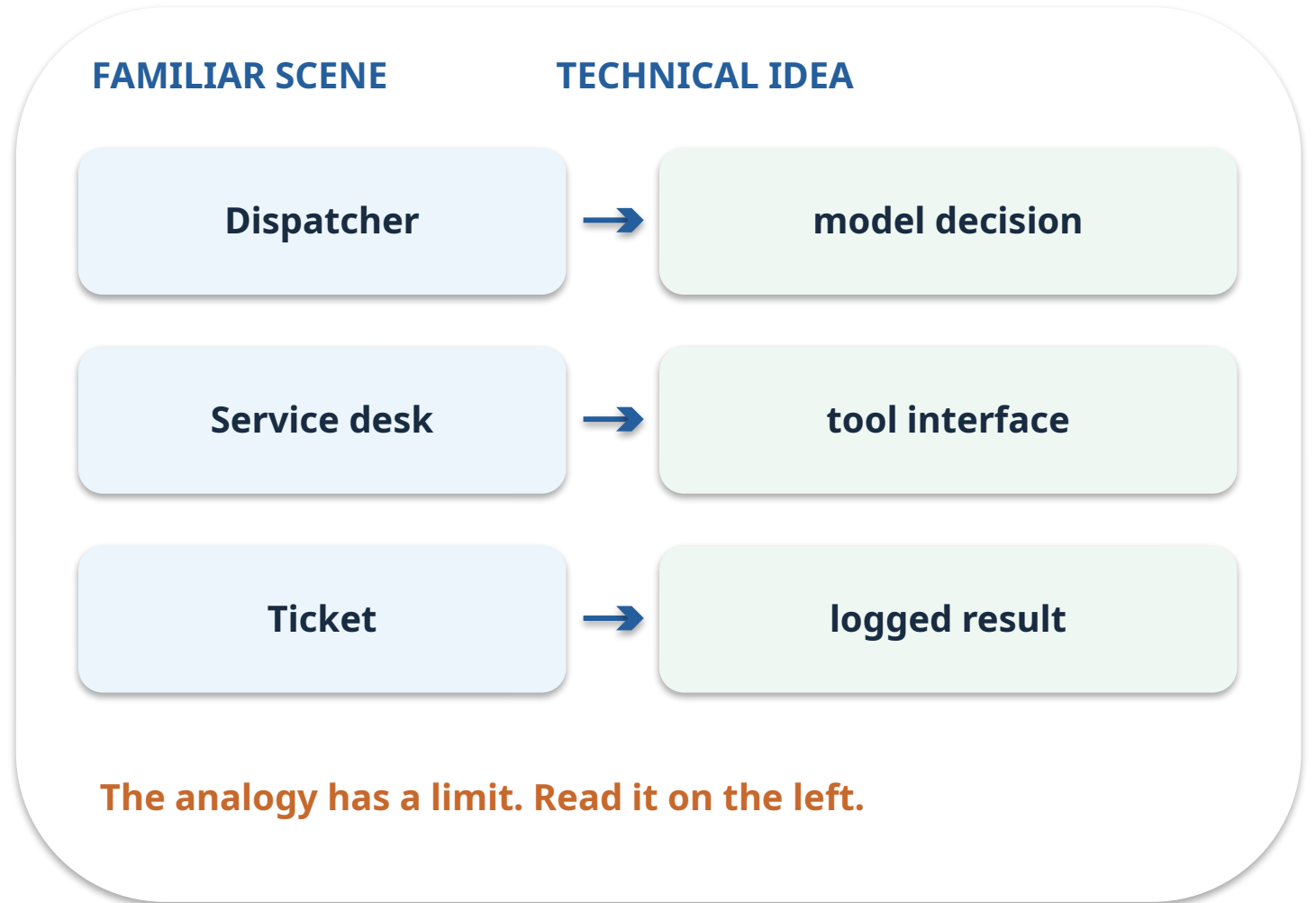
**What matches the technical process?**

# Agent · map the parts

## MAPPING / WEEK 08

Match each everyday part to the technical part.

Analogy limit: Agent choice is bounded by available tools and permissions; autonomy is limited.



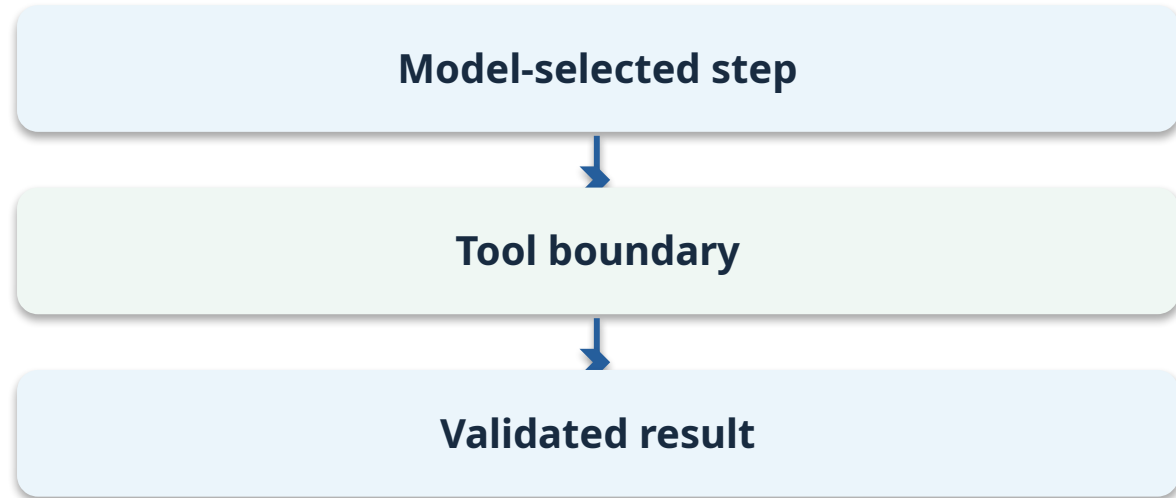
# Agent · how it works

## MECHANISM / WEEK 08

A model may choose a next action, within the permissions and limits supplied by the application.

Trace the arrows from left to right.

### FOLLOW THE DATA THROUGH EACH ARROW



# Agent · work one case

## WORKED / WEEK 08

The model proposes calculator;  
the interface validates €55  
before executing it.

Point to the evidence before  
giving your conclusion.

### GIVEN

**The model proposes calculator; the interface validates €55 before executing it.**

### QUESTION

**Can the model authorize a write by choosing a tool?**

### SUPPORTED ANSWER

**No. Authorization is enforced outside the model.**

# Agent · answer and explain

## REVEAL / WEEK 08

No. Authorization is enforced outside the model.

Tempting mistake: An agent can grant itself a new permission.

### TEST THE CLAIM AGAINST THE EVIDENCE

#### TEMPTING CLAIM

**An agent can grant itself a new permission.**

#### SUPPORTED EXPLANATION

**No. Authorization is enforced outside the model.**

Agent choice is bounded by available tools and permissions; autonomy is limited.

## 4. Predict · Approval boundary

### QUESTION / WEEK 08

What evidence shows a write was authorized?

Make a choice before we explain the diagram.

### VOTE FIRST · GIVE A REASON

**A An approval tied to the request and recorded action.**

**B A polite request is enough approval.**

**Which source or test could settle this?**

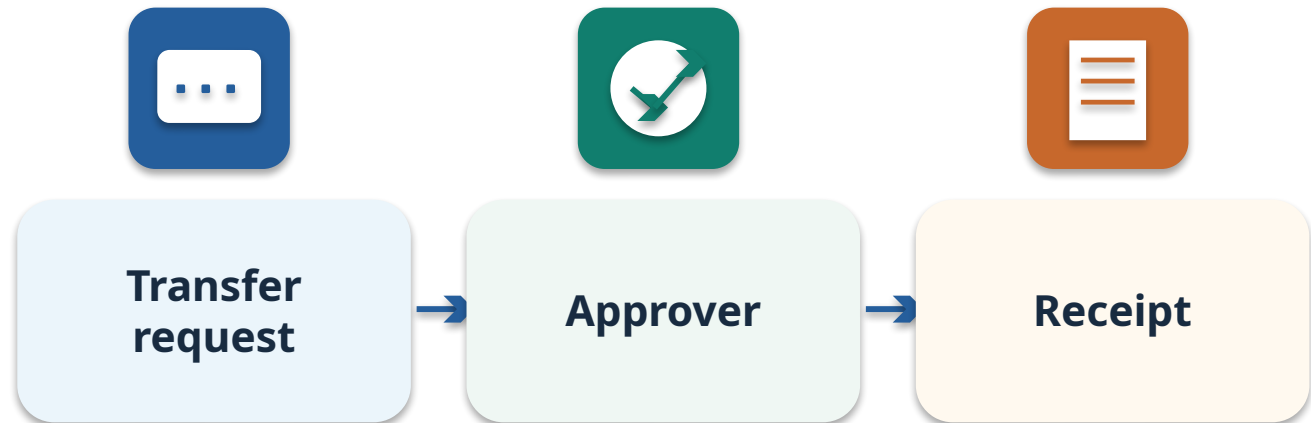
# Approval boundary · a familiar picture

## ANALOGY / WEEK 08

A bank transfer waits for an authorized person to approve it.

What matches our case, and where does the analogy stop?

**A bank transfer waits for an authorized person to approve it.**



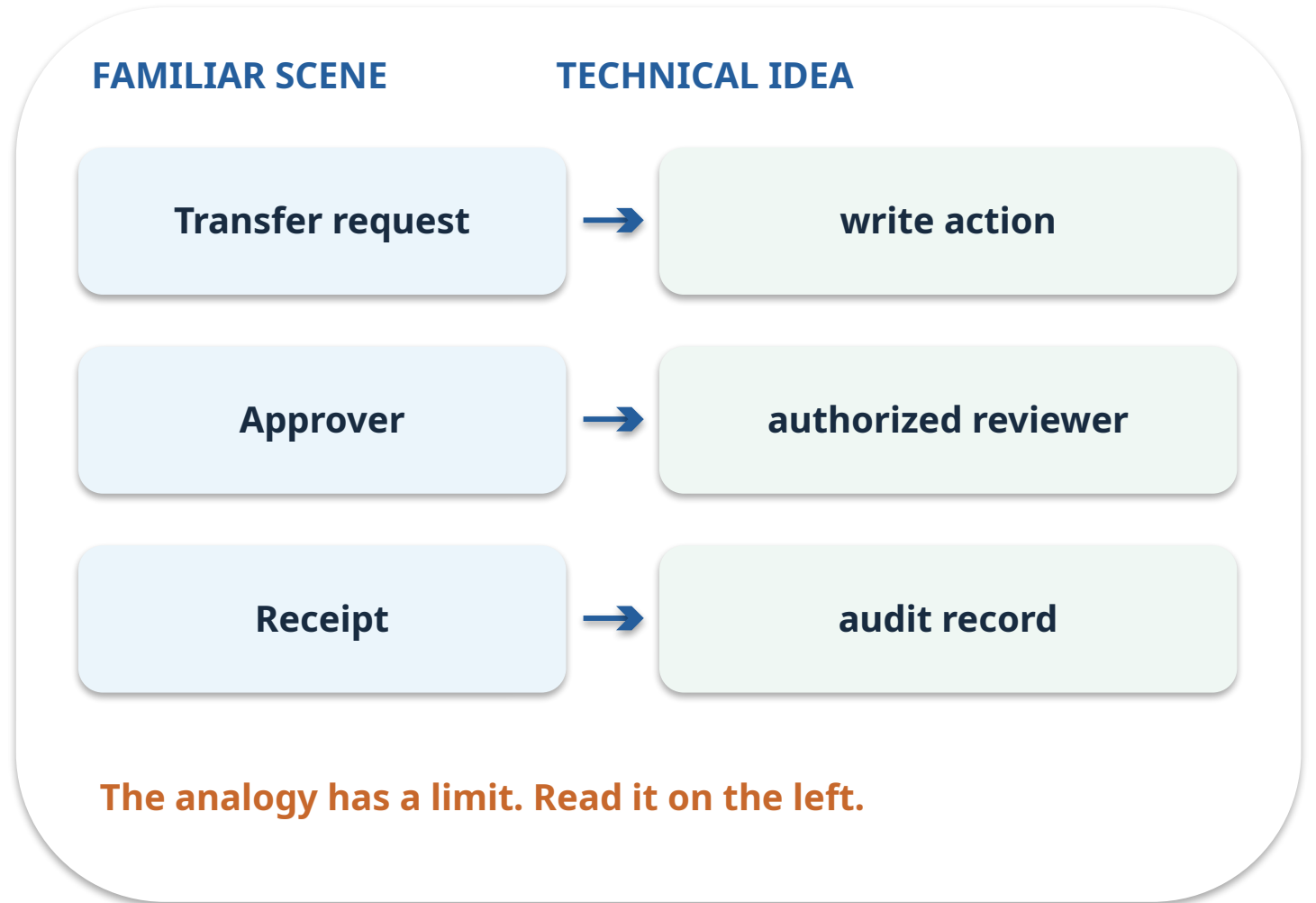
**What matches the technical process?**

# Approval boundary · map the parts

## MAPPING / WEEK 08

Match each everyday part to the technical part.

Analogy limit: Some read-only actions also need controls when they reveal restricted information.



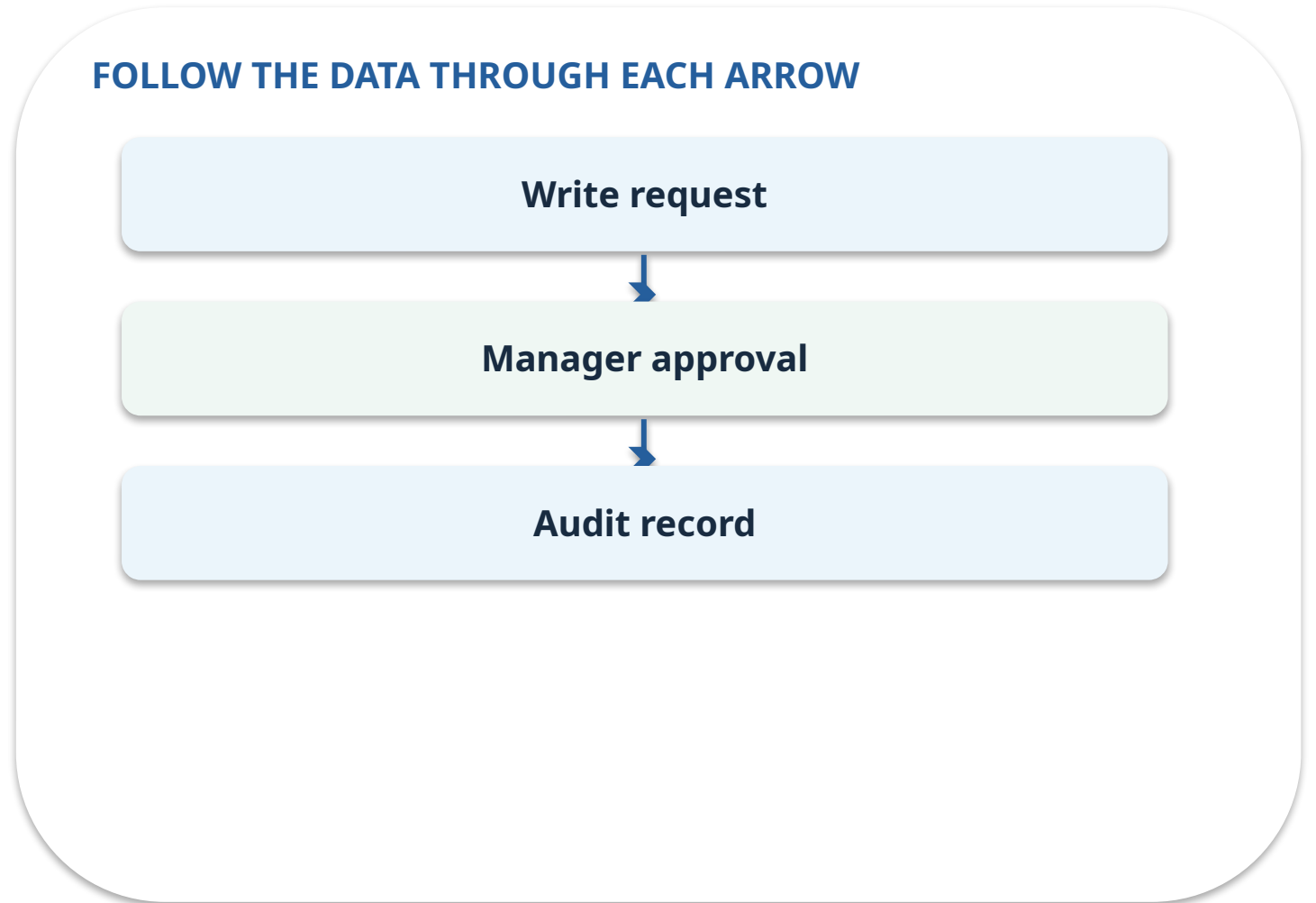
# Approval boundary · how it works

## MECHANISM / WEEK 08

A consequential write needs explicit review before execution and an audit record.

Trace the arrows from left to right.

### FOLLOW THE DATA THROUGH EACH ARROW



# Approval boundary · work one case

## WORKED / WEEK 08

Changing the cap requires an approval record; calculation of €40 does not change the record.

Point to the evidence before giving your conclusion.

### GIVEN

**Changing the cap requires an approval record; calculation of €40 does not change the record.**

### QUESTION

**What evidence shows a write was authorized?**

### SUPPORTED ANSWER

**An approval tied to the request and recorded action.**

# Approval boundary · answer and explain

## REVEAL / WEEK 08

An approval tied to the request and recorded action.

Tempting mistake: A polite request is enough approval.

### TEST THE CLAIM AGAINST THE EVIDENCE

#### TEMPTING CLAIM

**A polite request is enough approval.**

#### SUPPORTED EXPLANATION

**An approval tied to the request and recorded action.**

Some read-only actions also need controls when they reveal restricted information.

# Evidence trail: input → result

## WORKED / WEEK 08

Read-only tool returns €40.

A request to change the policy cap is denied until the authorized reviewer approves.

### GIVEN

**Policy cap is €40; eligible expense is €55.**

### QUESTION

**Read-only tool returns €40.**

### SUPPORTED ANSWER

**A request to change the policy cap is denied until the authorized reviewer approves.**

# Notebook: run the prepared example

## PRACTICE / WEEK 08

Implement a calculator and approval gate simulation.

Read the original output before editing.

1

**Call a read-only capped calculator**

2

**Reject a negative or non-numeric amount**

3

**Simulate an approval request without a real write**

4

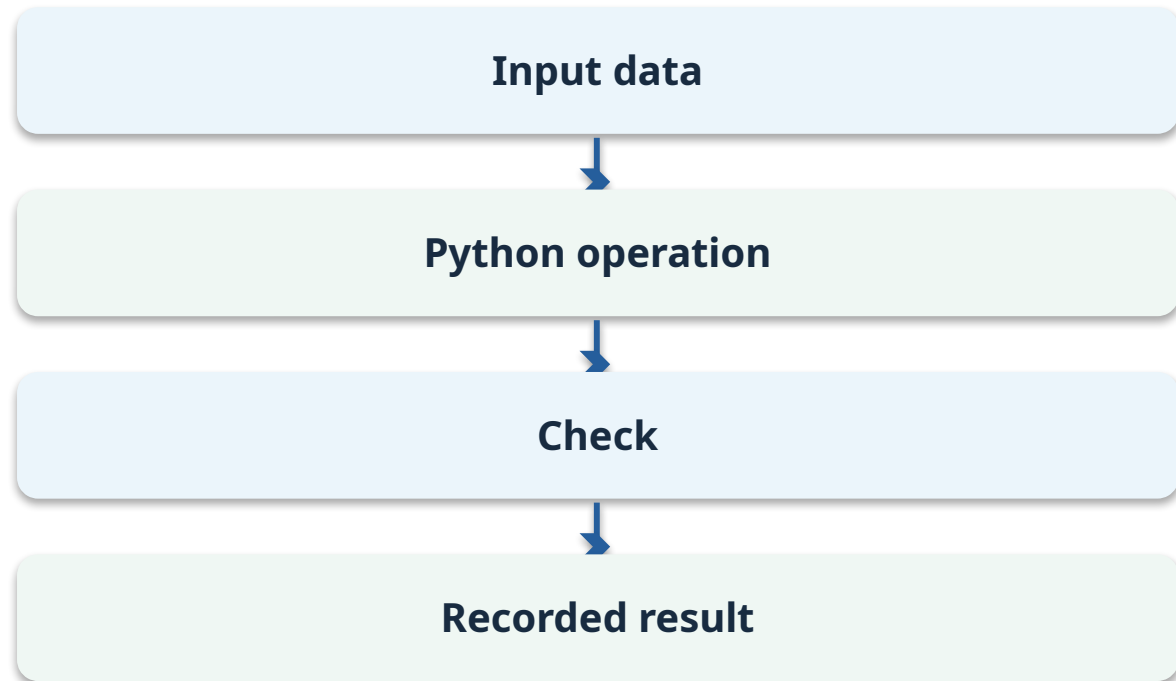
**Check that a denied request remains unchanged**

# Find the input and check in Python

## PRACTICE / WEEK 08

Locate the supplied input, the transformation, the checking line and the printed result.

### FOLLOW THE DATA THROUGH EACH ARROW



# Predict one controlled change

## PRACTICE / WEEK 08

Try a negative amount and confirm the validator blocks it.

What should change in the result? What stays fixed?

**Before: original case**

**After: one change**

**Keep all other inputs fixed.**

# Run, compare and record

PRACTICE / WEEK 08

Record the input, the one changed setting, the observed result and a reason for any difference.

|          |          |         |
|----------|----------|---------|
| Input    | Setting  | Result  |
| Original | baseline | record  |
| Changed  | one edit | compare |

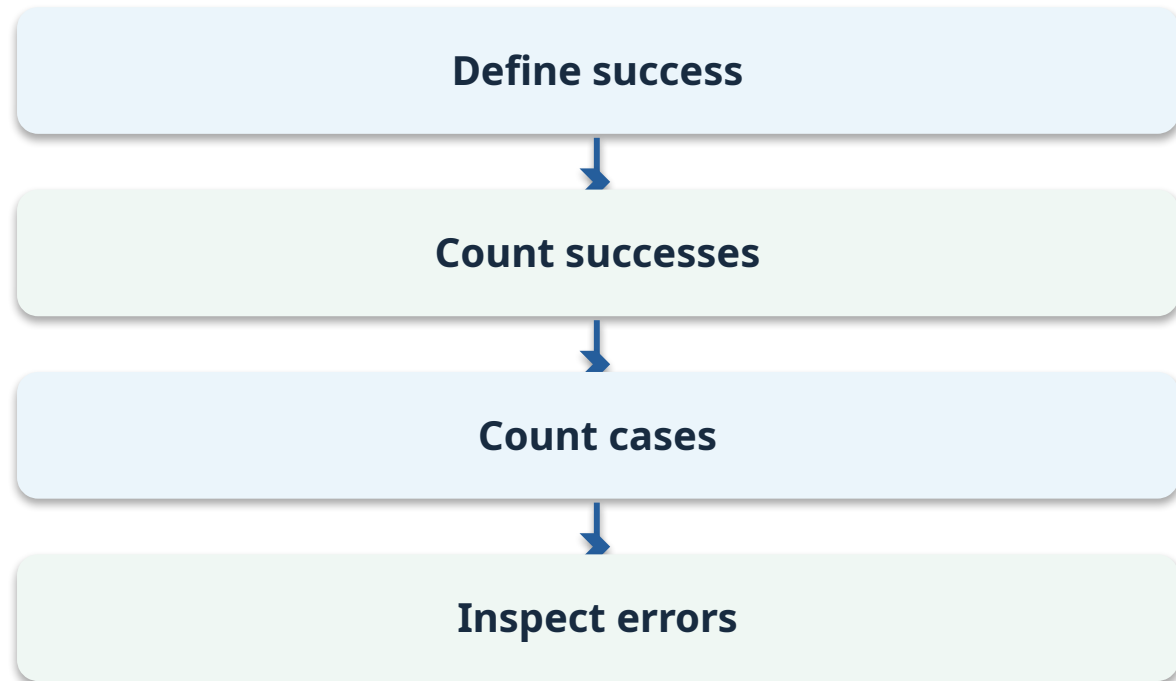
# Measure without hiding the denominator

## CHECK / WEEK 08

Unauthorized successful actions  
/ attempted unauthorized  
actions (target zero)

Write the numerator,  
denominator and one failure  
case.

### FOLLOW THE DATA THROUGH EACH ARROW



# Error detective: find the layer

## CHECK / WEEK 08

Plausible wrong claim: A tool call chosen by a model is automatically authorized.

Correct it using the case evidence.

### TEST THE CLAIM AGAINST THE EVIDENCE

#### TEMPTING CLAIM

**A tool call chosen by a model is automatically authorized.**

#### SUPPORTED EXPLANATION

**The application checks role, arguments and approval independently of the model text.**

Name a test that would catch this failure.

# Transfer to a new situation

## CHECK / WEEK 08

A tool receives amount –€5.  
What should happen before any calculation?

Explain which source or test would settle it.

### VOTE FIRST • GIVE A REASON

**A Use the visible evidence**

**B Flag missing or unsafe evidence**

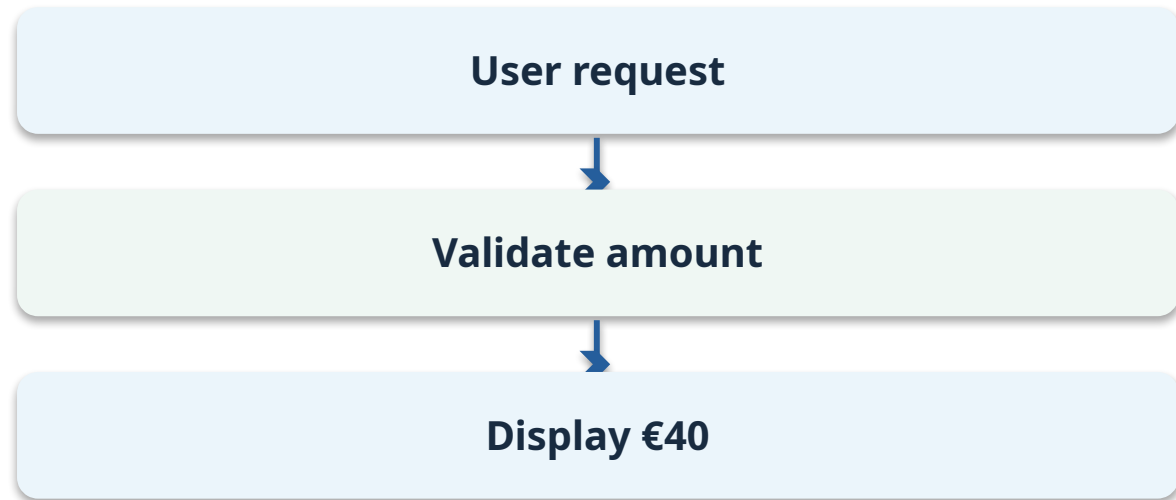
**Which source or test could settle this?**

# Explain it to a partner

## CHECK / WEEK 08

Without reading the slide, name the input, one processing step and the check. Your partner should ask “How do you know?”

### FOLLOW THE DATA THROUGH EACH ARROW



# Exam rehearsal · explain and apply

## CHECK / WEEK 08

What distinguishes a workflow from an agent?

Write a short answer, then compare with the guide.

1

**State the core idea**

2

**Use a case detail or calculation**

3

**Add a limitation**

4

**Compare with the Word guide**

# Exam rehearsal · explain and apply

## CHECK / WEEK 08

Calculate a reimbursement with eligible expense 55 and cap 40.

Write a short answer, then compare with the guide.

1

**State the core idea**

2

**Use a case detail or calculation**

3

**Add a limitation**

4

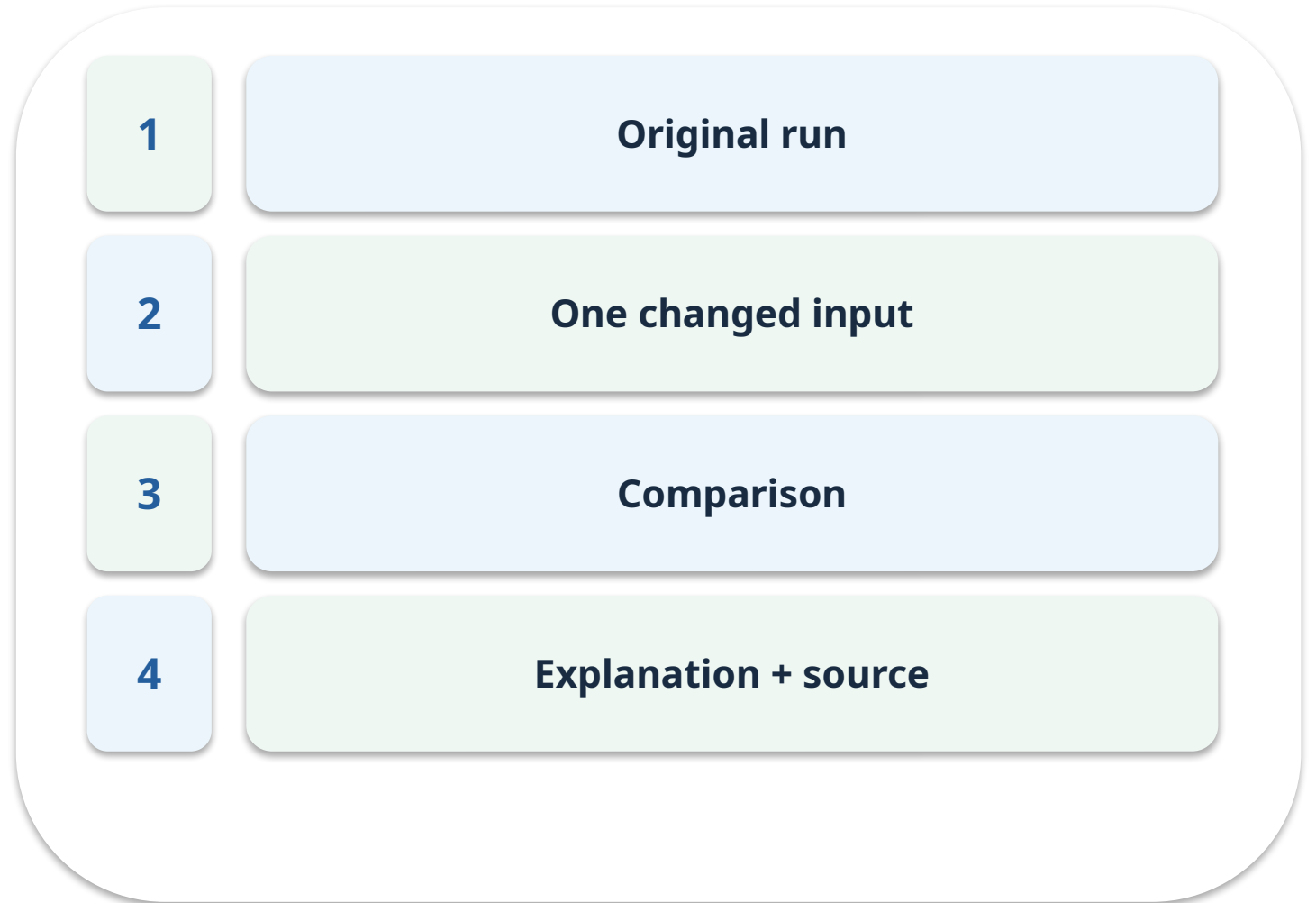
**Compare with the Word guide**

# What to hand in

## PRACTICE / WEEK 08

Tool result, blocked-request trace and approval explanation.

Include the one changed input and an error explanation.



# Three ideas to recall next week

## CHECK / WEEK 08

Validate a read-only tool; route write actions through human approval.

Read the linked official sources in the Word guide and redraw one diagram from memory.

1

**Tool**

2

**Workflow**

3

**Agent**

4

**Approval boundary**